

PRAKTIKUM BASIS DATA

MODUL 08

Scripting SQL (Create index, Create view, Analyze Table)



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

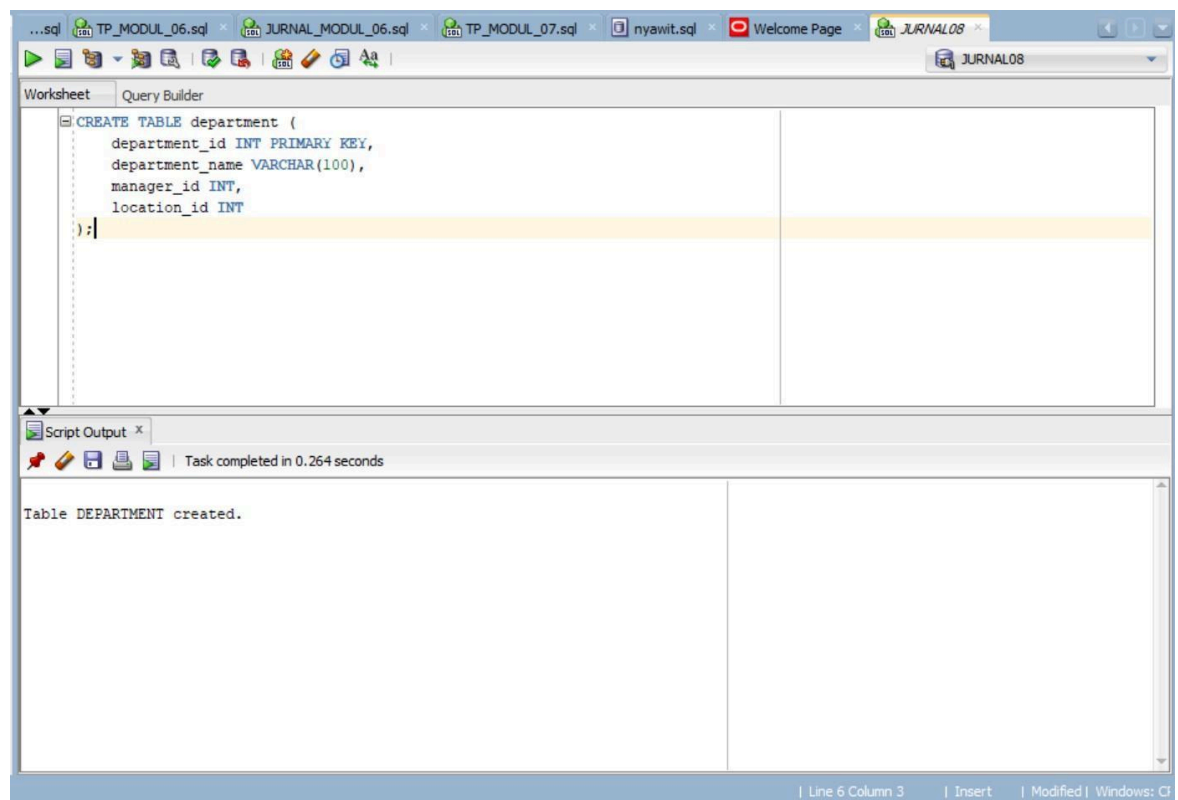
UNIVERSITAS TELKOM

2026

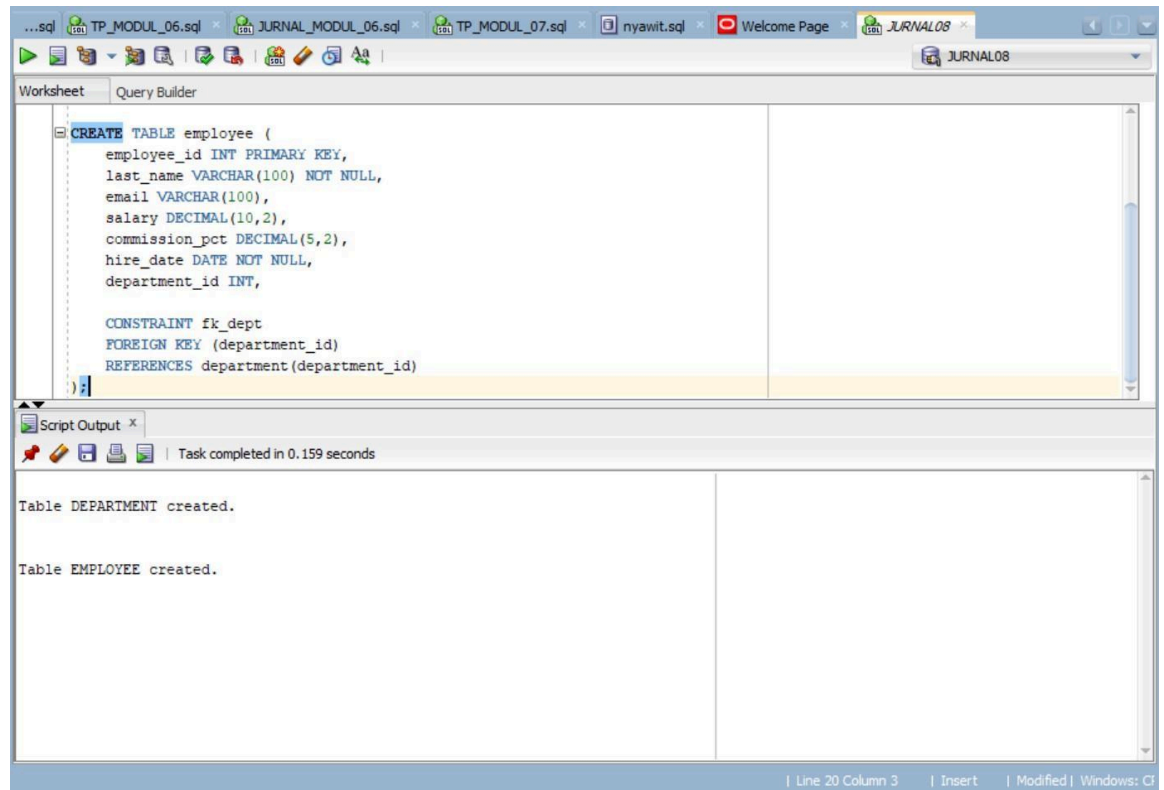
8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
 - a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
 - b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
 - c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

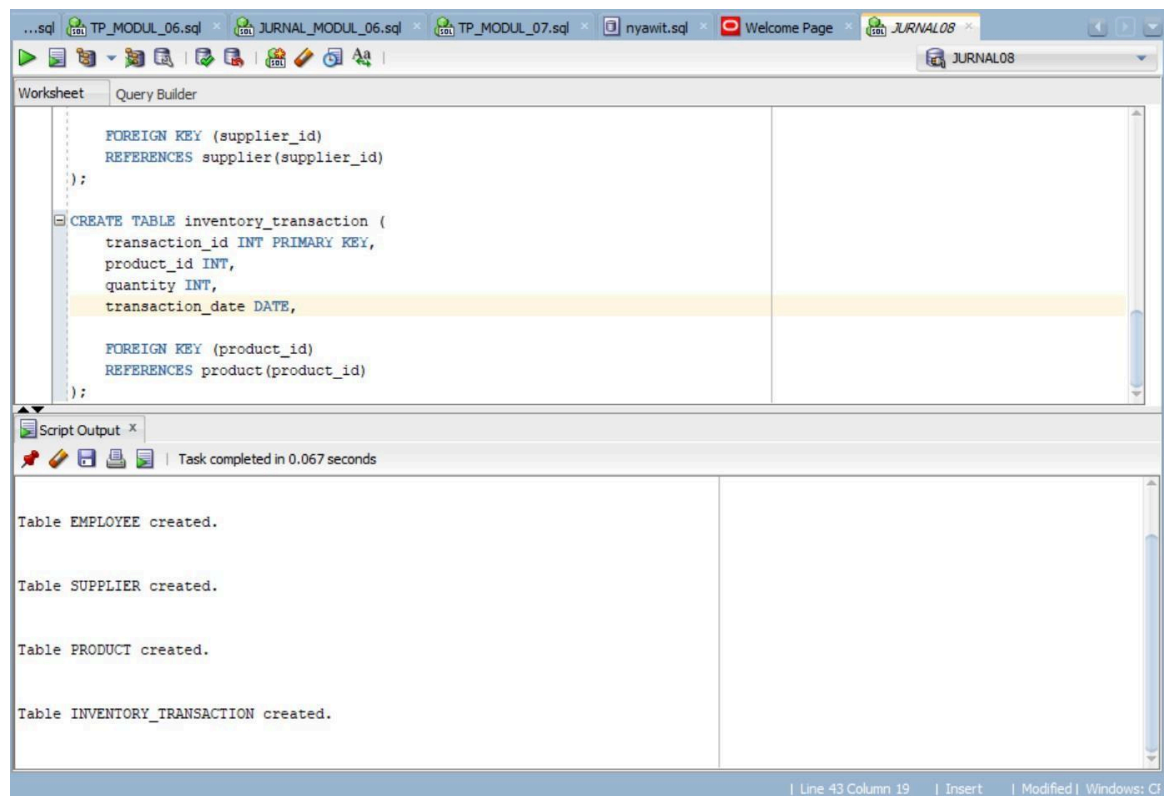
1. Desain Database



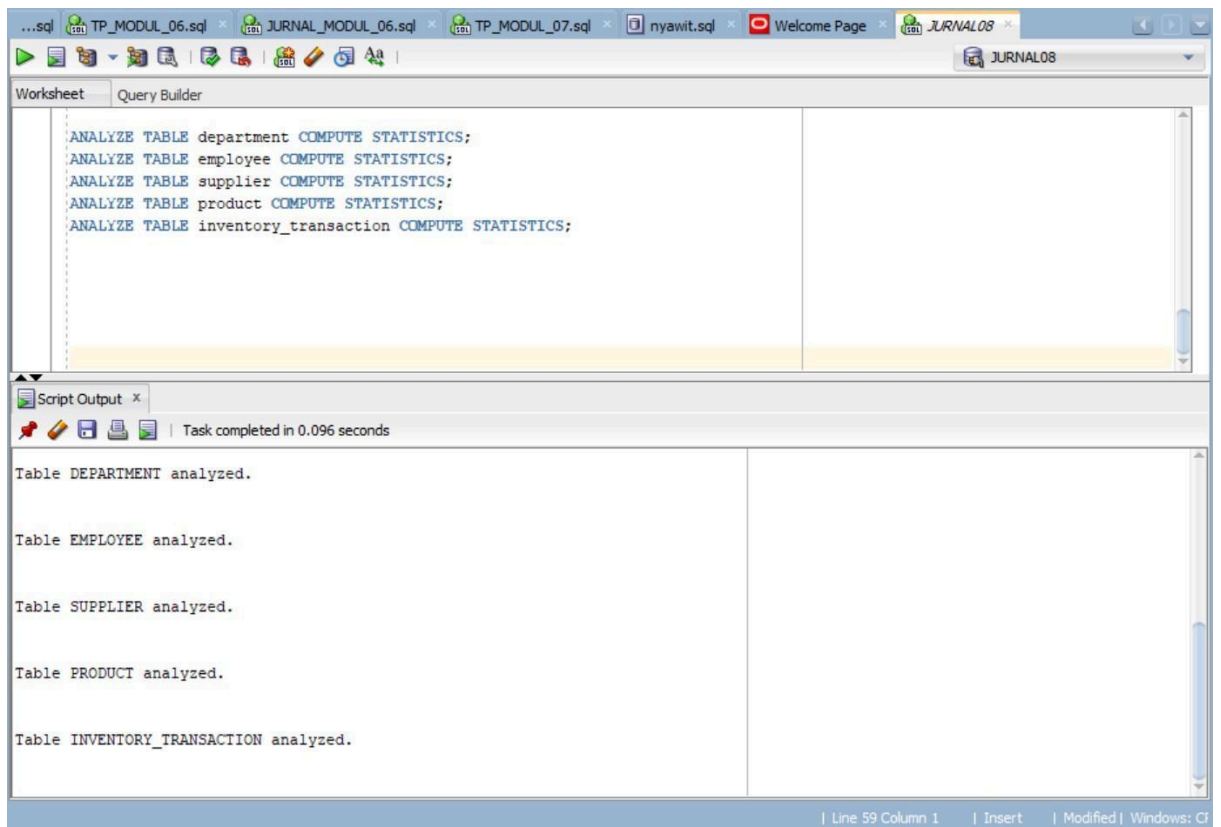
a.



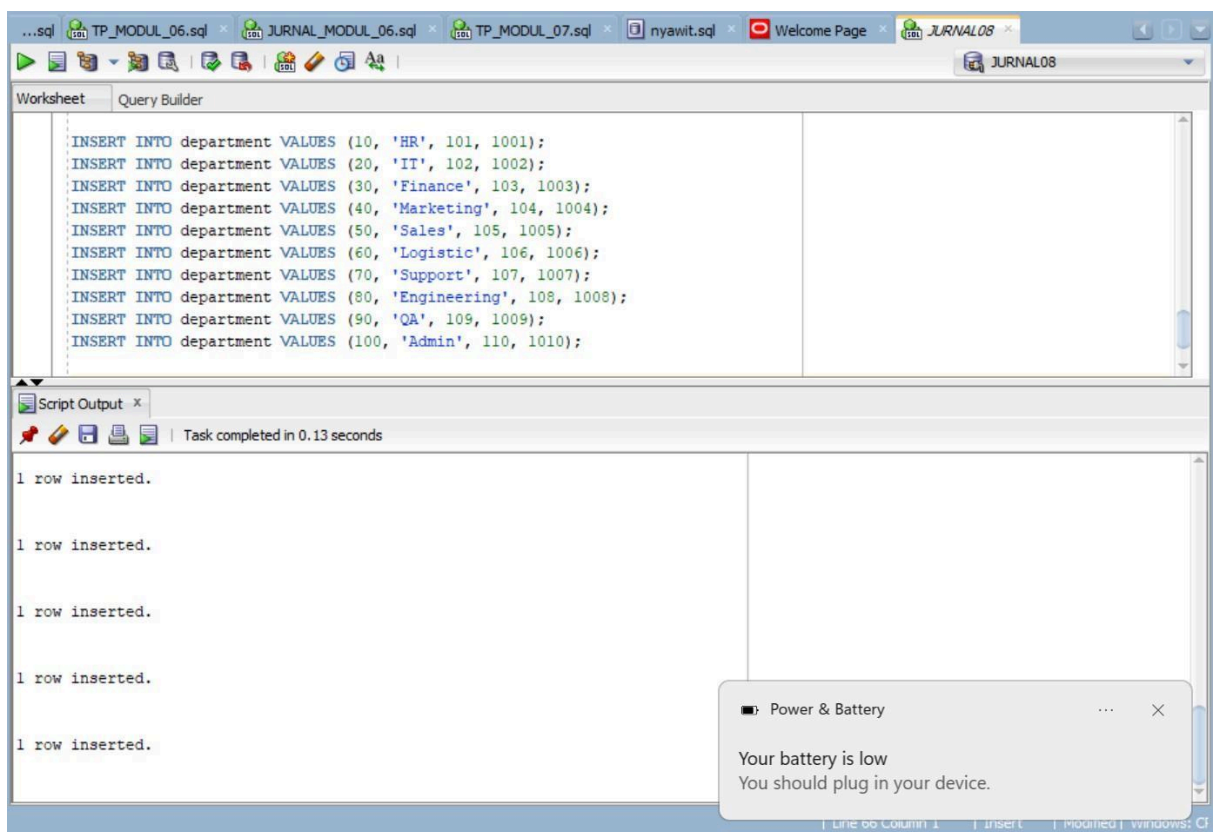
b.



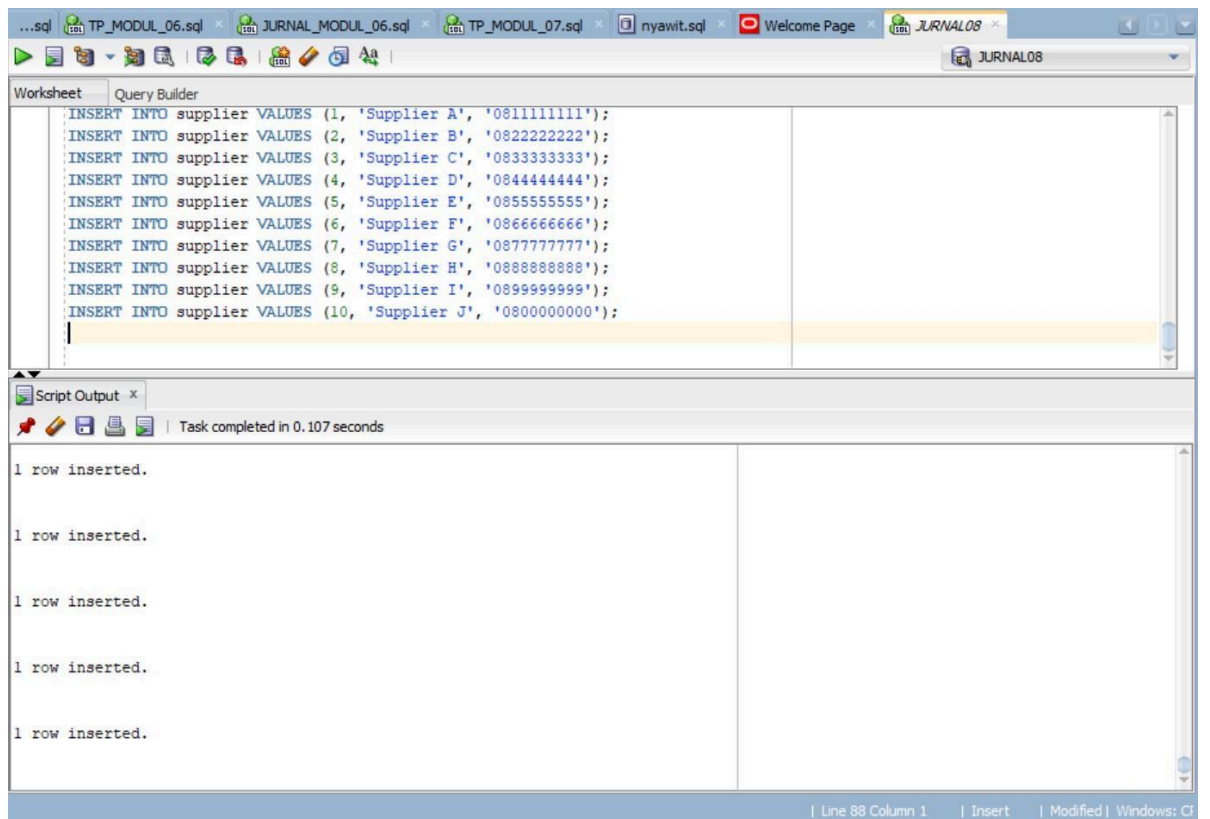
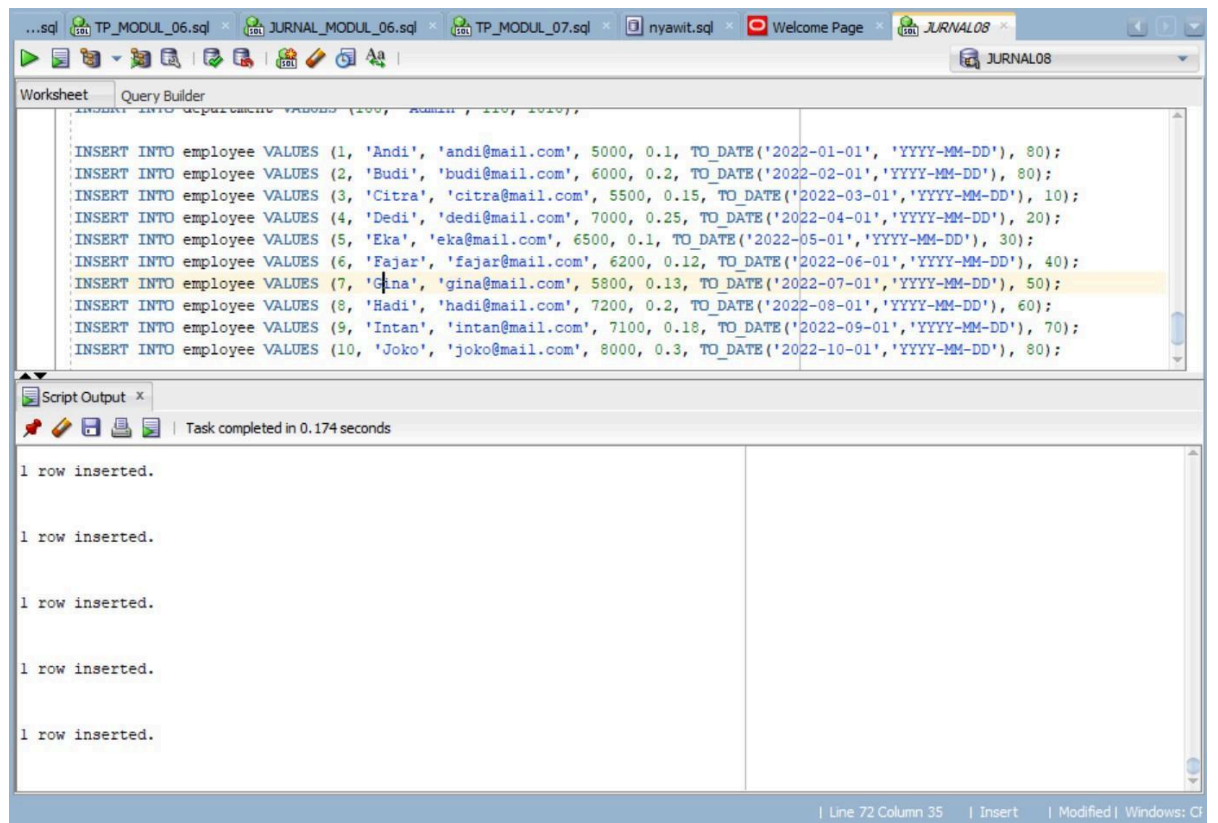
c.



2.



3.



...sql TP_MODUL_06.sql JURNAL_MODUL_06.sql TP_MODUL_07.sql nyawit.sql Welcome Page JURNAL08

Worksheet Query Builder

```
INSERT INTO product VALUES (1, 'Laptop', 10000000, 10, 1);
INSERT INTO product VALUES (2, 'Mouse', 200000, 50, 2);
INSERT INTO product VALUES (3, 'Keyboard', 300000, 40, 3);
INSERT INTO product VALUES (4, 'Monitor', 2000000, 20, 4);
INSERT INTO product VALUES (5, 'Printer', 1500000, 15, 5);
INSERT INTO product VALUES (6, 'Scanner', 1200000, 10, 6);
INSERT INTO product VALUES (7, 'Speaker', 500000, 25, 7);
INSERT INTO product VALUES (8, 'Webcam', 400000, 30, 8);
INSERT INTO product VALUES (9, 'Router', 800000, 18, 9);
INSERT INTO product VALUES (10, 'Flashdisk', 100000, 100, 10);
```

Script Output x

Task completed in 0.156 seconds

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

...sql TP_MODUL_06.sql JURNAL_MODUL_06.sql TP_MODUL_07.sql nyawit.sql Welcome Page JURNAL08

Worksheet Query Builder

```
INSERT INTO product VALUES (10, 'Flashdisk', 100000, 100, 10);
INSERT INTO inventory_transaction VALUES (1, 1, 5, TO_DATE ('2024-01-01', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (2, 2, 10, TO_DATE ('2024-01-02', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (3, 3, 8, TO_DATE ('2024-01-03', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (4, 4, 6, TO_DATE ('2024-01-04', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (5, 5, 7, TO_DATE ('2024-01-05', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (6, 6, 4, TO_DATE ('2024-01-06', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (7, 7, 9, TO_DATE ('2024-01-07', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (8, 8, 3, TO_DATE ('2024-01-08', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (9, 9, 2, TO_DATE ('2024-01-09', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (10, 10, 15, TO_DATE ('2024-01-10', 'YYYY-MM-DD'));
```

Script Output x

Task completed in 0.118 seconds

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

...sql | TP_MODUL_06.sql | JURNAL_MODUL_06.sql | TP_MODUL_07.sql | nyawit.sql | Welcome Page | JURNAL08

Worksheet | Query Builder

```
REFERENCES product(product_id)
);

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE product COMPUTE STATISTICS;
ANALYZE TABLE inventory_transaction COMPUTE STATISTICS;

INSERT INTO department VALUES (10, 'HR', 101, 1001);
INSERT INTO department VALUES (20, 'IT', 102, 1002);
INSERT INTO department VALUES (30, 'Finance', 103, 1003);
```

Script Output x

Task completed in 0.108 seconds

Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.

Table PRODUCT analyzed.

Table INVENTORY_TRANSACTION analyzed.

Line 53 Column 50 | Insert | Modified | Windows: C

4.

...sql | TP_MODUL_06.sql | JURNAL_MODUL_06.sql | TP_MODUL_07.sql | nyawit.sql | Welcome Page | JURNAL08

Worksheet | Query Builder

```
INSERT INTO inventory_transaction VALUES (8, 8, 3, TO_DATE ('2024-01-08', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (9, 9, 2, TO_DATE ('2024-01-09', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (10, 10, 15, TO_DATE ('2024-01-10', 'YYYY-MM-DD'));

CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;
```

Script Output x

Task completed in 0.082 seconds

Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.

Table PRODUCT analyzed.

Table INVENTORY_TRANSACTION analyzed.

View EMPVU80 created.

Line 119 Column 26 | Insert | Modified | Windows: C

5.

Worksheet Query Builder

```

INSERT INTO inventory_transaction VALUES (10, 10, 15, TO_DATE ('2024-01-10', 'YYYY-MM-DD'));

CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;

DESC empvu80;

```

Script Output x

Task completed in 0.678 seconds

Table PRODUCT analyzed.

Table INVENTORY_TRANSACTION analyzed.

View EMPVU80 created.

Name	Null?	Type
ID_NUMBER	NOT NULL	NUMBER(38)
NAME	NOT NULL	VARCHAR2(100)
SALARY		NUMBER(10,2)
DEPARTMENT_ID		NUMBER(38)

Line 121 Column 14 | Insert | Modified | Windows: CI

6.

Worksheet Query Builder

```

SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;

DESC empvu80;

CREATE INDEX idx_employee_dept
ON employee(department_id);

```

Script Output x

Task completed in 0.044 seconds

Table INVENTORY_TRANSACTION analyzed.

View EMPVU80 created.

Name	Null?	Type
ID_NUMBER	NOT NULL	NUMBER(38)
NAME	NOT NULL	VARCHAR2(100)
SALARY		NUMBER(10,2)
DEPARTMENT_ID		NUMBER(38)

Index IDX_EMPLOYEE_DEPT created.

Line 124 Column 28 | Insert | Modified | Windows: CI

7.